

Seshadri Naik Moode

PhD in Civil Engineering (UPC, July 2026)



✉ seshadri.naik.moode@upc.edu

☎ +34 671825396

📍 B1-006, Jordi Girona 1-3 Barcelona
08034 Spain

🌐 [linkedin.com/in/seshadri-naik-5002a2119](https://www.linkedin.com/in/seshadri-naik-5002a2119)

🆔 orcid.org/0000-0002-9651-7535

Summary

PhD in Civil Engineering (UPC BarcelonaTech, 2026) working at the interface of traffic flow theory, optimal control, and physics-based modelling. Developed an analytically string-stable car-following model with provable safety guarantees for CAV platoons and a hierarchical control framework integrating tactical and operational decisions across vehicle, maneuver, and freeway scales. Strengths in optimization, stability analysis, and constraint handling provide the mathematical foundations of hybrid, physics-informed traffic prediction. Proficient in Python and Aimsun.

Academic degrees

01/2022 – 07/2026
Barcelona, Spain

PhD in Civil Engineering

Universitat Politècnica de Catalunya, BarcelonaTech

Member of the Barcelona Innovative Transportation (BIT) [✉](#) research group, concentrating on platooning connected autonomous vehicles (CAVs) with an emphasis on microscopic modeling and management strategies, under the guidance of Prof. Francesc Soriguera [✉](#).

Academic stay: Research stay at KU Leuven Institute for Mobility, hosted by Prof. Claudio Roncoli (September 2025-January 2026)

09/2022 – Present

Urban Mobility

EIT Urban Mobility

Member of Doctoral Training Network funded by EIT Urban Mobility

2017 – 2019
Warangal, India

Masters in Transportation Engineering

National Institute of Technology, Warangal

Dissertation Title: Passenger Car Units for Urban Roads under Mixed Traffic Conditions. Under the supervision of Dr. Arpan Mehar, Assistant Professor, NIT Warangal.
Tools: PTV Vissim

2013 – 2017
RK Valley, India

Bachelors in Civil Engineering

Rajiv Gandhi University of Knowledge Technologies (RGUKT), RK Valley

Publications and Conference Proceedings

Soriguera, F., **Moode, S.N.**, Martinez-Diaz, M. (2026) A safe space-gap model for the platooning of connected autonomous vehicles in freeways: stability and safety analysis. Submitted to *IEEE Transactions on Intelligent Transportation Systems*. (Under Review. Manuscript available upon request).

Moode, S.N., Soriguera, F., Martinez-Diaz, M. (2026) Freeway lane management strategies in mixed traffic with platoons of connected and automated vehicles. Submitted to *Physica A: Statistical Mechanics and its Applications*. (Under Review. Manuscript available upon request).

Zheng, X., Soriguera, F., Xie, C., & **Moode, S. N.** (2025, June). An interaction-aware predictive control framework with adaptive gap regulation in connected and automated vehicle platoons. In *2025 IEEE Intelligent Vehicles Symposium (IV)* (pp. 1674-1680). IEEE. 10.1109/IV64158.2025.11097812 [✉](#)

Oriol, L., Martínez-Díaz, M., **Moode, S. N.**, Sala, M., & Soriguera, F. (2024). Platooning of connected automated vehicles on freeways: a microsimulation approach. *Transportmetrica B: Transport Dynamics*, 12(1), 2360543. <https://doi.org/10.1080/21680566.2024.2360543> [✉](#)

Moode, S. N., Soriguera, F., Zheng, X., & Martínez Díaz, M. (2024). Exploring safety in platoons of connected autonomous vehicles: investigating emergency braking conditions and collision severity. In Proceedings of the 12th Symposium of the European Association for Research in Transportation (hEART). https://transportor.epfl.ch/heart/2024/abstracts/hEART_2024_paper_7796.pdf 

Moode, S. N., & Soriguera, F. (2023). Microscopic modeling of connected autonomous vehicles platooning: stability and safety analysis. *Transportation Research Procedia*, 71, 260-267. <https://doi.org/10.1016/j.trpro.2023.11.083> 

Moode, S. N., & Soriguera, F. (2023). Platooning of connected autonomous vehicles in freeway traffic: state of the art. *Transportation Research Procedia*, 71, 268-275. <https://doi.org/10.1016/j.trpro.2023.11.084> 

Moode, S.N., Soriguera, F., Martinez-Diaz, M. (2026) Energy and emissions of freeway lane management strategies with platoons of connected autonomous vehicles. Working Paper/Manuscript in preparation (manuscript available upon request)

Moode, S. N., Roncoli, C., and Soriguera, F. (2026) Hierarchical optimal control for lane management of CAVs in non-cooperative mixed traffic. Working Paper/Manuscript in preparation (manuscript available upon request)

Conferences

06/2023 Tenerife, Spain	Congreso de Ingenieria del Transporte (CIT 2023) <i>Microscopic modeling of connected autonomous vehicles platooning: stability and safety analysis. (Oral Presentation)</i>
09/2023 Santander, Spain	Euro Working Group on Transportation (EWGT) <i>String-stable car-following model algorithm for platoons of connected autonomous vehicles. (Oral Presentation)</i>
06/2024 Espoo, Finland	12th Symposium of the European Association for Research in Transportation. (HEART) <i>Exploring safety in platoons of connected autonomous vehicles: investigating emergency braking conditions and collision severity. (Oral Presentation)</i>
09/2024 Heraklion, Greece	Conference on Emerging Technologies (TRC-30) <i>Definition of strategies to optimize the platooning of connected automated vehicles on freeways. (Poster Presentation)</i>
09/2024 Heraklion, Greece	The 5th Symposium on Management of Future Motorway and Urban Traffic Systems (MFTS) <i>Examining Emergency Braking in Connected Autonomous Vehicle Platooning: Collision Risk and Severity Analysis. (Oral Presentation)</i>
01/2025 Washington, DC, USA	104th Transportation Research Board Annual Meeting (TRB) <i>Impact of lane management strategies of connected autonomous vehicle platoons on traffic flow: a simulation study. (Poster Presentation)</i>
06/2025 Zaragoza, Spain	Congreso de Ingenieria del Transporte (CIT 2025) <i>Lane Management Strategies for CAV Platoons: A Comparative Study of Mixed and Dedicated Lanes. (Oral Presentation)</i>
06/2025 Okinawa, Japan	The 12th Triennial Symposium on Transportation Analysis conference (TRISTAN XII) <i>Lane Management Strategies to Enhance Traffic Performance in Mixed Traffic Environments with Platoons of Connected Autonomous Vehicles (Oral Presentation)</i>

Skills

- **Simulation tools:** Aimsun, PTV
- **Programming & Scientific Computing:** Python, Aimsun Scripting
- **Modelling:** Microscopic traffic modelling, CAV behaviour modelling
- **Domains:** Connected & Automated Vehicles, traffic flow theory, safety analysis
- **Methods:** Optimal control, dynamic programming, control barrier functions, stability analysis, and optimization.

Awards

06/2023	Young Researcher Award CIT 2023 <i>PTV Group Award for Young Researchers in Mobility and Accessibility in Transport</i> for an article on microscopic modeling of CAV platooning.
2022	AGAUR PhD Grant <i>Agència de Gestió d'Ajuts Universitaris i de Recerca. Generalitat de Catalunya</i>

Languages

English C1	Spanish B1/B2	Telugu
Hindi	Catalan A2	

Doctoral Training Network Annual Forum

2022, 2023, 2024 Editions in Barcelona, Spain; Espoo, Finland; and Ghent, Belgium respectively.

Research Impact

- Research contributes to understanding **safety implications of CAV deployment in real-world traffic**
- Supports development of **methodologies for modelling and management of automated vehicle systems**
- Findings relevant for **traffic management strategies and future regulatory frameworks**

Research projects involved

09/2020 – 06/2021 Hyderabad, India	TIHAN IIT Hyderabad <i>Designed India's first autonomous vehicle testbed using AutoCAD, which set a national benchmark for research and development in the field of autonomous navigation.</i> • Led research on vehicle autonomy levels and traffic adaptation, providing critical insights that helped enhance the implementation of autonomous navigation in mixed Indian traffic conditions. • Collaborated on safety system evaluations and qualitative studies, contributing to the development of reliable and context-aware navigation and collision warning systems, which improved overall safety for autonomous vehicles.
---------------------------------------	--

Teaching activities

Madanapalle, India	Assistant Professor <i>SVTM, Civil Engineering Department</i> • Used a variety of learning modalities and support materials to facilitate the learning process and accentuate presentations, including visual and social learning modalities. • Supported weekly transportation and fluid mechanics lab sessions, contributing to student engagement, comprehension, and learning objectives.
Barcelona, Spain	Graduate Student Instructor <i>Universitat Politècnica de Catalunya</i> Assisted Prof. Francesc Soriguera in his teaching courses, particularly in Traffic and Operations courses. Co-supervised two master's theses with Prof. Soriguera, from problem definition through simulation design to defense.

Courses and training activities

12/2018 NIT Warangal, India	GIAN course on "Sustainable Mobility and Transportation: Innovative approaches and solutions"
08/2018 NIT Warangal, India	GIAN course on "Transportation and the Environment"

05/2018 NIT Warangal, India	Participated in 5th Colloquium on Transportation systems engineering and Management
01/2018 NIT Warangal, India	GIAN course on "Decision Tools for Selection and Evaluation of Transportation Projects"
NIT Warangal, India	Participated in Five-day Faculty development program on Road safety issues and challenges

Internships

Hyderabad, India	M.Tech Internship <i>VR TECHNICHE Consultants Pvt.Ltd.</i> Preparation of project reports & schedules for award on long term fee collection for toll plazas.
------------------	---

References

Francesc Soriguera ✉, *Professor*, Universitat Politècnica de Catalunya - BarcelonaTech
francesc.soriguera@upc.edu

Claudio Roncoli ✉, *Associate Professor*, KU Leuven
claudio.roncoli@kuleuven.be

Margarita Martinez-Diaz ✉, *Associate Professor*, Universitat Politècnica de Catalunya - BarcelonaTech
margarita.martinez.diaz@upc.edu